Survival Models Tutorial 1: Solutions

1. We have

$$S_x(t) = \Pr(T > x + t \mid T > x) = \frac{\Pr(T > x + t, T > x)}{\Pr(T > x)} = \frac{\Pr(T > x + t)}{\Pr(T > x)} = \frac{S(x + t)}{S(x)}.$$

2. Since

$${}_tq_x = F_x(t) = \int_{s=0}^{t} f_x(s)\, ds$$

we just need to show that

$$f_x(t) = {}_tp_x\mu_{x+t}.$$

We have

$$\begin{aligned}
f_x(t) &= \frac{\partial}{\partial t}F_x(t) = \frac{\partial}{\partial t}\left(1 - S_x(t)\right) \\
&= -\frac{\partial}{\partial t}S_x(t) \\
&= -\lim_{h\to 0+}\frac{S_x(t+h) - S_x(t)}{h} \\
&= \lim_{h\to 0+}\frac{1}{h}\left(\frac{S(x+t)}{S(x)} - \frac{S(x+t+h)}{S(x)}\right) \\
&= \frac{S(x+t)}{S(x)}\lim_{h\to 0+}\frac{1}{h}\left(\frac{S(x+t)}{S(x+t)} - \frac{S(x+t+h)}{S(x+t)}\right) \\
&= \frac{S(x+t)}{S(x)}\lim_{h\to 0+}\frac{1}{h}\left(\frac{F(x+t+h) - F(x+t)}{S(x+t)}\right) \\
&= {}_tp_x\lim_{h\to 0+}\frac{1}{h}(F_{x+t}(h)) \\
&= {}_tp_x\lim_{h\to 0+}\frac{1}{h}({}_hq_{x+t}) \\
&= {}_tp_x\mu_{x+t}.
\end{aligned}$$

Interpretation: The chance of dying sometime in x to $x+t$ is found by adding up over s all the chances of surviving from x to $x+s$ (this is ${}_sp_x$) and then dying in $x+s$ to $x+s+ds$ (this is $\mu_{x+s}\,ds$).

3. We know that ${}_tq_x = \Pr(T < x + t|T > x) = F_x(t)$ and ${}_tp_x = S_x(t)$ so from Q2 we have

$$\begin{aligned}
F_x(t) &= \int_0^t {}_sp_x\mu_{x+s}\, ds \\
\Rightarrow f_x(s) &= S_x(s)\mu_{x+s} \\
\Rightarrow \int_0^t \frac{f_x(s)}{S_x(s)}\, ds &= \int_0^t \mu_{x+s}\, ds \\
\Rightarrow -\int_0^t \frac{\partial}{\partial s}\log S_x(s)\, ds &= \int_0^t \mu_{x+s}\, ds \\
\Rightarrow \log S_x(t) &= -\int_0^t \mu_{x+s}\, ds
\end{aligned}$$

and the result follows.

4. (a) If the force of mortality is a constant μ over an interval $[x, x+t]$ then ${}_tp_x = \exp(-\mu t)$. Applying this result we have
(i) ${}_2p_{70} = \exp(-\mu_0) \times \exp(-\mu_1) = \exp[-(\mu_0 + \mu_1)]$ and
(ii) ${}_2p_{70} = \exp[-(\mu_0 + \mu_1)/2 \times 2] = \exp[-(\mu_0 + \mu_1)]$.
(iii) The general rule is: to evaluate ${}_np_x$ where n is an integer and the force of mortality is piecewise constant over single years we may replace the forces of mortality by their mean acting over all n years.

(b) (i) Using the result above, the mean force of mortality is

$$\frac{(3833 + \ldots 5609)}{5} \times 10^{-5} = 0.046742$$

so ${}_5p_{70} = \exp(-0.046742 \times 5) = 0.792$.
In actual fact, the tables assume that the value of the force of mortality applies at an exact age, and that this is approximately the mean of the interval ± 0.5, i.e. we can assume the value of μ_{70} approximates the value from 69.5 up to 70.5. Using this we have

$${}_5p_{70} = \exp\left(-[0.5 \times 3833 + 4198 + \ldots + 5609 + 0.5 \times 6123] \times 10^{-5}\right) = 0.783.$$

This value of 0.783 agrees with

$$l_{75}/l_{70} = 53266/68055$$

and

$$(1 - q_{70}) \times (1 - q_{71}) \times \ldots \times (1 - q_{74}).$$

(ii) The mean force of mortality is

$$\frac{(1342 + \ldots 1949)}{5} \times 10^{-5} = 0.016288$$

so ${}_5p_{65} = \exp(-0.016288 \times 5) = 0.922$.
(iii) Pr(Both survive 5 years) $= 0.792 \times 0.922 = 0.730$.
(iv) Pr(Exactly one survives 5 years) $= 0.792 \times 0.078 + 0.208 \times 0.922 = 0.25$.

5. (a) We use the standard formula

$$\begin{aligned} {}_3p_{70} &= \exp\left\{-\int_0^3 \mu_{70+t}\, dt\right\} \\ &= \exp\left\{-\left(\int_0^1 0.04\, dt + \int_1^2 0.045\, dt + \int_2^3 0.05\, dt\right)\right\} \\ &= e^{-(0.04+0.045+0.05)} = e^{-0.135} = 0.874 \end{aligned}$$

and so ${}_3q_{70} = 0.126$.

(b) The probability that two such lives survive and one dies is

$$3\ {}_3p_{70}^2\ {}_3q_{70} = 3 \times 0.874^2 \times 0.126 = 0.289.$$